Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/30/2024
PI Name:	Lundblad, Nathan Ph.D.		
Project Title:	Microgravity Dynamics of Bubble-Geometry Bose-Einstein Condensates		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	04240-6018	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	09/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2025 per D. Griffin/NASA-HQ (Ed., 1/20/25). NOTE: Contract number changed to JPL 1699891 per J. Callas/JPL (Ed., 8/29/24). NOTE: End date changed to 9/27/2024 per U. Israelsson/JPL (Ed., 10/20/21) NOTE: End date changed to 3/31/2022 per B. Carpenter/NASA HQ (Ed., 1/4/2021) NOTE: New end date is 10/30/2020 per JPL (Ed., 5/21/19)		
Key Personnel Changes/Previous PI:	New postdoc Dr. Jean-Baptiste Gerent hired in March 2024.		
COI Name (Institution):	Aveline, David Ph.D. (Jet Propulsion Laboratory) Vishveshwara, Smitha Ph.D. (University of Illinois at Urbana-Champaign)		
Grant/Contract No.:	JPL 1699891 ; JPL 1502172		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Notions of geometry, topology, and dimensionality have directed the historical development of quantum-gas physics. With a toolbox of forces used to confine, guide, and excite Bose-Einstein condensates (BEC) or degenerate Fermi gases (DFG), physicists have used quantum gases to test fundamental ideas in quantum theory, statistical mechanics, and in recent years notions of strongly-correlated many-body physics from the condensed-matter world. We propose a specific program to explore a trapping geometry for quantum gases that is both tantalizing theoretically and prohibitively difficult to attain terrestrially: a quantum gas in a bubble geometry, i.e., a trap formed by a spherical or ellipsoidal shell structure, confining a 2D quantum gas to the surface of an experimentally-controlled topologically-connected “bubble.” The physics of a quantum gas confined to such a surface has not been explored terrestrially due to the limitations of gravitational sag; interesting work has certainly been done with gases confined to the lower regions of bubble potentials, but the fully symmetric situation has yet to be explored. The low-energy excitations of such a system are unexplored, and notions of vortex creation and behavior as well as Kosterlitz-Thouless physics are tantalizing aims as well. The solid-state modeling goals of the optical-lattice physics community are also fundamentally connected to the system, as the canonical Mott-insulator/superfluid transition features superfluid shells isolated between insulating regions. The central method to reach the sought-after bubble-geometry BEC or DFG is that of rf or microwave dressing of the bare trapping potentials provided by the Cold Atom Laboratory (CAL) “chip trap.” Radiofrequency dressing has been used conceptually through "rf-knife" evaporative cooling, but more recently through explicit construction of adiabatic potentials for interferometry, and shell-trap construction for both thermal and quantum gases. The proposed work is a window into a physical regime that is quite difficult to achieve terrestrially due to trap distortion; given the advantages of a microgravity environment, NASA CAL is uniquely positioned to realize the physics goals of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work, while focused on the fundamental physics of ultracold atoms and not directly connected to human life, has a similar impact to life on Earth as that of all fundamental physics; it broadens our understanding of the physical world and helps us further cement our collective picture of quantum mechanics as "the way the world works." It explores the limits of how large Bose-Einstein condensates can be made, and to what extent the gravity-well of terrestrial labs render certain investigations difficult or impossible. The observations made aboard CAL through this project are a clear demonstration that physical insight can sometimes require microgravity facilities to be fully developed, and that spaceborne atomic physics experiments can be valuable contributions to our collective scientific efforts.
Task Progress:	Task Progress: The 2023-2024 period of this work was focused on planning for the SM3B generation of the NASA Cold Atom Laboratory (CAL) data acquisition. [Ed. Note: SM3B refers to Phase 2 ("SM3") of the CAL operation.] Dr. Lundblad was central driver of this work in this period, together with our partner at Jet Propulsion Laboratory (JPL), David Aveline, who, in addition to service as Co-Investigator, was our primary liaison to experimental operations. Theory Co-Investigators Smitha Vishveshwara and student Brendan Rhyno provided regular insight and support, especially in regard to computational modeling of observed phenomena. Results of our research activities were presented at several conferences and international workshops, including the 2023 Bad Honnef ultracold atomic physics summer school (https://). [Ed. Note: See Bibliography.]
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Abstracts for Journals and Proceedings	Lundblad N. "Ultracold atomic physics in microgravity." Ultracold Atoms and Molecules Bad Honnef Physics School Workshop, Bad Honnef, Germany, August 8-12, 2023. Abstracts. Ultracold Atoms and Molecules Bad Honnef Physics School Workshop, Bad Honnef, Germany, August 8-12, 2023. https://www.dpg-physik.de/veranstaltungen/2023/lundblad-1.pdf , Aug-2023
Articles in Peer-reviewed Journals	Raudonis M, Roura A, Meister M, Lotz C, Overmeyer L, Herrmann S, Gierse A, Lämmerzahl C, Bigelow NP, Lachmann M, Piest B. "Microgravity facilities for cold atom experiments. " Quantum Sci Technol. 2023 Aug 10;8(4):044001. https://doi.org/10.1088/2058-9565/ace1a3 , Aug-2023
Articles in Peer-reviewed Journals	Elliott ER, Aveline DC, Bigelow NP, Boegel P, Botsi S, Charron E, d’Incao JP, Engels P, Estrampes T, Gaaloul N, Kellogg JR. "Quantum gas mixtures and dual-species atom interferometry in space. " Nature. 2023 Nov 16;623(7987):502-8. https://doi.org/10.1038/s41586-023-06645-w , Nov-2023